

NewJeans Stem Player

Improved Project Concept Document

This revised document restructures the original PDF into a clearer product specification while preserving the original concept, visual identity, intended functionality, and technical detail. It also incorporates two confirmed design decisions: the player uses stem-summed playback, and the top-right menu is dedicated to song navigation only.

The project is an interactive NewJeans stem player that lets users browse releases, choose a song, follow timed lyrics, identify the currently singing member, and manipulate separated audio stems in real time.



Figure 1. Original home page concept: a release-browsing interface arranged from newest to oldest.

Core Strengths

Area	Strength
Concept	A focused fan-oriented stem player rather than a generic audio player.
Visual Identity	Bright blue, yellow, pink, stars, rounded panels, member portraits, and playful UI language create a cohesive NewJeans-inspired style.
Interaction Model	The flow from release selection to song selection to stem playback is clear and natural.
Feature Depth	Timed lyrics, language switching, member indicators, waveforms, mute, solo, volume, and timeline scrubbing give the concept real product depth.

1. Product Overview

The NewJeans Stem Player is designed as a playful music exploration tool. Users start by browsing NewJeans releases, select an EP, album, or single, choose a song from that release, then open the main stem player workspace.

The experience is built around three major layers: release browsing, song selection, and stem-based playback. The strongest part of the concept is that it combines fan identity with music-analysis functionality. It does not only play a track; it lets users see who is singing, follow lyrics in multiple formats, and isolate parts of the arrangement.

Primary User Flow

Home Page -> Release Selection -> Song Selection Page -> Stem Player Page

This flow should remain unchanged. It is simple, sequential, and easy to understand.

Assumptions

The following assumptions are used where the original PDF did not provide complete implementation context:

- The project is either a private prototype or will use legally cleared audio, artwork, lyrics, and stem files.
- The stem files for each song are time-aligned and can be played simultaneously without drifting.
- The language selector uses English, Korean, and romanized lyric data stored with timestamps.
- The member indicator system is powered by timed metadata rather than automatic vocal detection.

2. Improved Page Structure

The original PDF explains the screens in sequence, but the writing reads like screenshot captions. A stronger structure is to describe each page using the same product-spec format: purpose, user actions, UI elements, and interaction behavior.

Recommended PDF structure:

1. Project Overview
2. User Flow
3. Home Page
4. Song Selection Page
5. Stem Player Page
6. UX and Design Notes
7. Feature Enhancements
8. Technical Implementation Notes
9. Final Evaluation

3. Home Page - Revised Specification

The home page displays all NewJeans releases in reverse chronological order, from newest to oldest. Users can scroll through the release list and select an EP, album, or single to open its song selection page.

Each release should appear as a clickable card containing the cover artwork, release title, release year, release type, and song count. The select/play button on each card should act as the main call-to-action.

The bottom-left icon and bottom-right link connect to the creator's Twitter/X account. These links should remain secondary to the main music-navigation controls.

Home Page UI Elements

Element	Purpose	Recommended Detail
Release cards	Allow the user to choose an EP, album, or single.	Include artwork, title, year, release type, and song count.
Scroll area	Lets the user browse the full release list.	Keep newest releases at the top.
Card action button	Opens the selected release.	Label clearly with Select, Open, or View Songs.
Creator links	Connect to Twitter/X.	Keep small and visually secondary.

Improved Home Page Copy

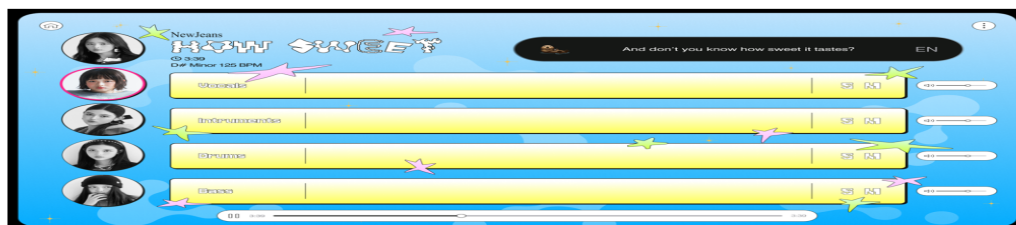
Users begin on the home page, where all NewJeans releases are listed from newest to oldest. Each release card includes the cover image, title, release year, release type, and number of songs. Selecting a release opens the song selection page for that specific EP, album, or single.



This is the song selection page. This page will allow you to choose whichever song you would like from the EP/album using the pink arrows.

The title will show the song name, as well as the release year and the type of release it is.

The home icon in the top left corner takes you back to the home page.



This is the stem player page. This page will allow you to view and play the song of your choice.

Figure 2. Original song selection and stem player screens. The top screen functions like a song carousel; the bottom screen is the main mixer workspace.

4. Song Selection Page - Revised Specification

The song selection page lets users choose a song from the selected release. It uses a carousel-style interface: the pink left and right arrow buttons move between songs, while the main artwork and title update to show the currently selected song.

The centered Select button confirms the current song and opens the stem player page. The home icon in the top-left corner returns users to the main release page.

Song Selection UI Elements

Element	Purpose	Recommended Detail
Song artwork	Shows the current selected song or release artwork.	Keep the current selection visually dominant.
Pink arrows	Cycle between songs in the selected release.	Add hover labels: Previous Song and Next Song.
Song title and metadata	Clarifies what the user is selecting.	Show song name, year, and release type.
Select button	Opens the stem player.	Rename to Open Stem Player or Play This Song for stronger clarity.
Home icon	Returns to the home page.	Increase size slightly or add hover label.

Improved Song Selection Copy

This page uses a carousel-style selector. The pink arrows allow users to cycle through songs from the selected release, while the Select button opens the currently displayed song in the stem player.

Navigation Rule

The song selection page should only handle choosing songs from the selected release. Deeper playback controls, lyric display, stem mixing, and song metadata belong on the stem player page.

5. Stem Player Page - Revised Specification

The stem player page is the main interactive workspace. It allows users to play the selected song, follow timed lyrics, identify which member is singing, and control individual audio stems.

The top-left section displays the song title, song length, musical key, and BPM. To the right, a timed lyrics box shows the current lyric line in sync with the music. The language selector allows users to switch between English, Korean, and romanized lyrics.

Beside the lyric line, an emoji indicates which member is currently singing. The member portraits on the left update in real time, with the active singer highlighted and inactive members grayed out.

Member Indicator System

Indicator	Member / Meaning
Teddy bear	Minji
Otter	Hanni
Dog	Danielle
Cat	Haerin
Hamster	Hyein
Bunny	Full group or unison vocal section

Mixer Workspace

The center of the page contains the Interactive Multi-Track Audio Mixer Workspace. Each row represents a separate audio stem, including vocals, instruments, drums, and bass. Every row includes a waveform display, synchronized playhead, mute and solo controls, and an individual volume slider.

At the bottom, the main audio timeline scrubber shows the song's playback position and allows users to jump to different points in the track. The scrubber moves in sync with the playhead across all stem rows.

Confirmed Song Navigation Menu

The top-right icon should not be a general settings menu. It should open a song navigation menu only. This keeps the interface cleaner and avoids turning the menu into a miscellaneous control drawer.

Song Navigation Menu:

- Change Song
- Back to Song Selection
- Previous Song
- Next Song
- Current Release
- Current Song

Revised wording: The icon in the top-right corner opens the song navigation menu. This menu lets users switch to another song from the same EP, album, or single, move to the previous or next song, or return to the song selection page.

6. Confirmed Stem Playback Model

Use Model B: Stem-Summed Playback. This is the correct model for a real stem player because it allows mute, solo, and volume controls to affect the actual mix.

Default playback: Vocals + Instruments + Drums + Bass = Full mix

The player should use the separated stems as the main playback source. By default, all stems play together at the same time to recreate the full song mix. Each stem row controls its own audio file.

Stem Row Controls

Control	Function
Volume slider	Adjusts the level of that individual stem.
Mute	Removes that stem from the mix.
Solo	Isolates one or more selected stems.
Playhead	Moves in sync across all stems.
Waveform	Visualizes that specific stem.

Master File Role

The master file should not be the default playback source. It should only be used as a fallback if stem files are unavailable, still loading, or unsupported for a specific song.

This avoids a functional contradiction: if the master file plays by default, the individual stem volume sliders cannot meaningfully control the full mix. With stem-summed playback, every mixer control has a real audible effect.

Improved Stem Playback Copy

The stem player uses separated audio stems as the main playback source. By default, all stems play together in sync to recreate the full song mix. Each row controls one stem category, such as vocals, instruments, drums, or bass. Users can adjust the volume of each stem, mute individual stems, or solo one or more stems to isolate specific parts of the song. All stems share the same playback timeline and synchronized playhead, so changes happen in real time while the song plays. A master audio file may be used as a fallback only if stem playback is unavailable.

7. UX and Design Feedback

Navigation

The user flow is strong, but the navigation hierarchy should be more explicit. Home, song selection, song switching, and playback controls should remain visually distinct.

Current Element	Issue	Improvement
Home icon	Small and easy to miss.	Increase size or add hover label: Home.
Top-right icon	Originally too broad as settings.	Use only as Song Navigation.
Pink arrows	Clear visually but unlabeled.	Add hover labels: Previous Song and Next Song.
Select button	Functional but generic.	Rename to Open Stem Player or Play This Song.
Creator links	Useful but secondary.	Keep smaller than app navigation controls.

Layout Clarity

The stem player has the highest feature density. The interface should use a clear hierarchy: song info first, timed lyrics and vocalist indicator second, stem mixer third, transport controls fourth, secondary navigation last.

The lyric box should receive stronger contrast and more spacing because it is one of the project's most distinctive features.

Accessibility

Area	Recommendation
Member highlights	Use color plus border plus label. Do not rely on color alone.
Mute/Solo buttons	Use readable text labels or obvious active states.
Volume sliders	Show value feedback, such as a percentage or level indicator.
Language selector	Use full labels at least in the menu: English, Korean, Romanized.
Contrast	Increase contrast between yellow panels and light text.
Touch targets	Make arrows, icons, and sliders large enough for mobile use.

Interaction Feedback

Every major control should confirm its state. Active mute should look muted, active solo should look selected, volume changes should show level feedback, and loading states should appear when audio, lyrics, or artwork are not ready.

Mobile Layout

On mobile, the mixer should stack vertically. A practical order is: song info, lyrics and vocalist indicator, stem rows, then sticky playback controls at the bottom. Member portraits can collapse into smaller circular icons.

8. Feature Suggestions

These enhancements strengthen the project without changing its core identity. They keep the focus on fan listening, vocal tracking, and stem exploration.

Feature	Description	Why It Helps
Stem Presets	Add preset buttons such as Vocals Only, Instrumental, Rhythm Section, Karaoke Mode, and Production Mode.	Gives users instant useful actions without requiring them to understand mixer logic first.
Member Vocal Timeline	Add a small timeline showing which member sings at each section.	Turns the member indicator system into a vocal arrangement map.
Lyrics Click-to-Jump	Let users click a lyric line to jump to its timestamp.	Useful for close listening, translation, and replaying favorite moments.
A/B Loop Tool	Add Set A, Set B, Loop Section, and Clear Loop controls.	Makes the player useful for studying hooks, vocals, basslines, or drum grooves.
Waveform Zoom	Allow Full Song, Section View, and Bar/Beat View.	Makes the waveform more useful for detailed audio analysis.
Section Markers	Label song structure: Intro, Verse, Pre-Chorus, Chorus, Bridge, Outro.	Improves navigation and makes the player feel more music-aware.
Bookmarks	Let users save timestamped notes.	Allows fans to mark vocal entrances, drops, harmonies, or favorite moments.
Fallback States	Add states for lyrics unavailable, stem unavailable, artwork loading, and failed audio load.	Makes the product feel realistic and robust.

Recommended Priority Order

1. Stem presets
2. Lyrics click-to-jump
3. Section markers
4. A/B loop tool
5. Member vocal timeline
6. Bookmarks
7. Waveform zoom
8. Fallback states

9. Technical and Implementation Notes

Core Systems

System	Role
Web Audio API	Stem playback, gain control, mute, solo, and synchronized audio routing.
Wavesurfer.js	Waveform rendering and visual playback tracking.
Tone.js	Timing, transport logic, and scheduling if more precise sequencing is needed.
JSON metadata	Stores song info, lyric timestamps, language variants, and member indicators.
React or Next.js	Front-end application structure.
Tailwind CSS	Fast styling and responsive layout.
Zustand or Redux	Shared playback state, selected song, active stems, lyric state, and navigation state.

Recommended Song Metadata Structure

```
{
  "id": "how-sweet",
  "title": "How Sweet",
  "release": "How Sweet",
  "year": 2024,
  "releaseType": "EP",
  "duration": "3:39",
  "key": "D# Minor",
  "bpm": 125,
  "artwork": "/artwork/how-sweet.jpg",
  "master": "/audio/how-sweet/master.mp3",
  "stems": {
    "vocals": "/audio/how-sweet/vocals.mp3",
    "instruments": "/audio/how-sweet/instruments.mp3",
    "drums": "/audio/how-sweet/drums.mp3",
    "bass": "/audio/how-sweet/bass.mp3"
  },
  "lyrics": [
    {
      "start": 12.4,
      "end": 16.8,
      "text_en": "And don't you know how sweet it tastes?",
      "text_ko": "...",
      "text_rom": "...",
      "member": "hanni",
      "indicator": "otter"
    }
  ]
}
```

Playback Architecture

All stems must load from the same zero point and share the same transport state. Play, pause, seek, loop, mute, solo, and volume changes should update centrally so every stem remains locked to the same

timeline.

When the user seeks with the bottom scrubber, every stem should jump to the same timestamp. When a stem is muted, its gain should be set to zero rather than stopping that audio source independently. This reduces sync problems.

Copyright and Licensing Note

If the project is public-facing, it needs legally cleared audio, stem files, lyrics, and artwork. If it is a private prototype, the PDF should still state that assumption. Recommended note: This concept assumes that all audio, artwork, lyrics, and stem files are legally cleared for use or are being used only for private prototype demonstration.

10. Final Evaluation

The project has a strong core identity: a playful fan-facing NewJeans music explorer that combines release browsing, song selection, stem playback, timed lyrics, and member-based vocal tracking. The visual direction is cohesive and memorable. The feature set is more ambitious than a normal audio player, which gives the project a clear reason to exist.

Strongest Aspects

- Strong concept
- Clear page-to-page user flow
- Good fan-oriented visual identity
- Useful stem mixer system
- Timed lyrics plus member indicators add personality
- Release and song browsing make the experience feel complete

Biggest Opportunities

- Clarify all stem playback logic around stem-summed playback
- Keep the top-right menu dedicated to song navigation only
- Make controls larger and more clearly labeled
- Improve accessibility through contrast, labels, and non-color active states
- Strengthen mobile layout
- Add production/listening features such as presets, looping, section markers, and lyric click-to-jump
- Address copyright and licensing assumptions clearly

Final Verdict

The PDF is now stronger as a product concept document. The biggest functional improvement is the switch to stem-summed playback: all stems play together by default, and the user controls the mix directly. The biggest UX improvement is reducing the top-right menu to song navigation only. Together, these changes make the project cleaner, more technically coherent, and easier to understand.